

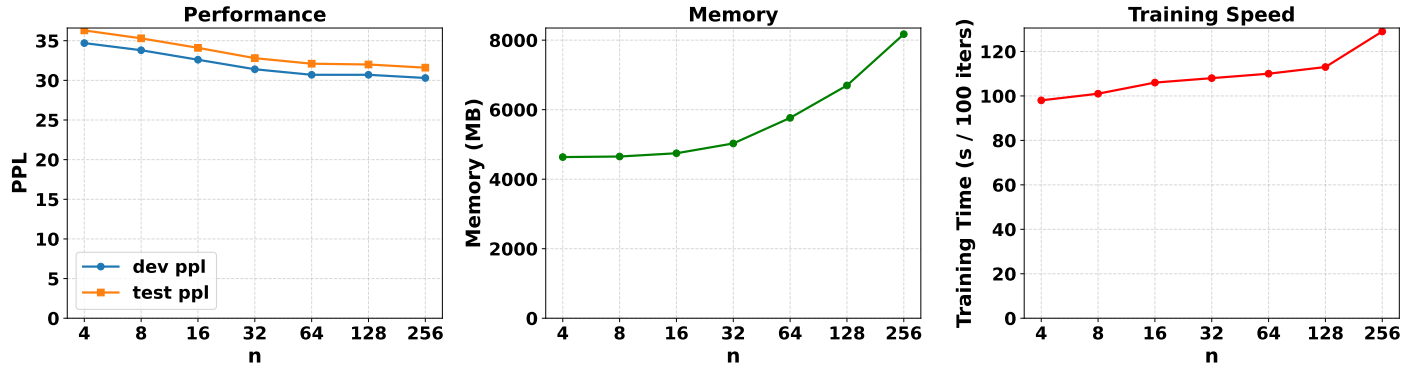


Figure 1: Performance, memory consumption, and training speed of Flexformer_n in the autoregressive language modeling experiments under different values of the sampling frequency hyperparameter n . On large-scale models and datasets, increasing n improves both the fitting and generalization abilities of the model, while also incurring higher computational cost. In the main experiments, we set $n = 64$.

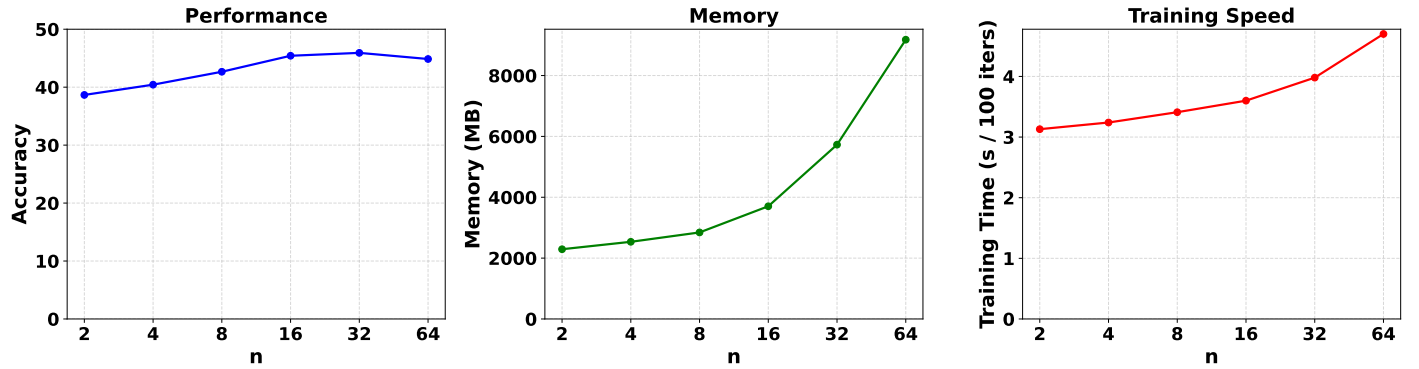


Figure 2: Performance, memory consumption, and training speed of Flexformer_n on the Image dataset for Long Sequence Classification under different values of the sampling frequency hyperparameter n . Both memory usage and training time increase with larger n . On the Image dataset, increasing n improves the model’s generalization performance when n is small; however, as n becomes larger, it increases computational cost and may even lead to overfitting. In the main experiments, we set $n = 16$.

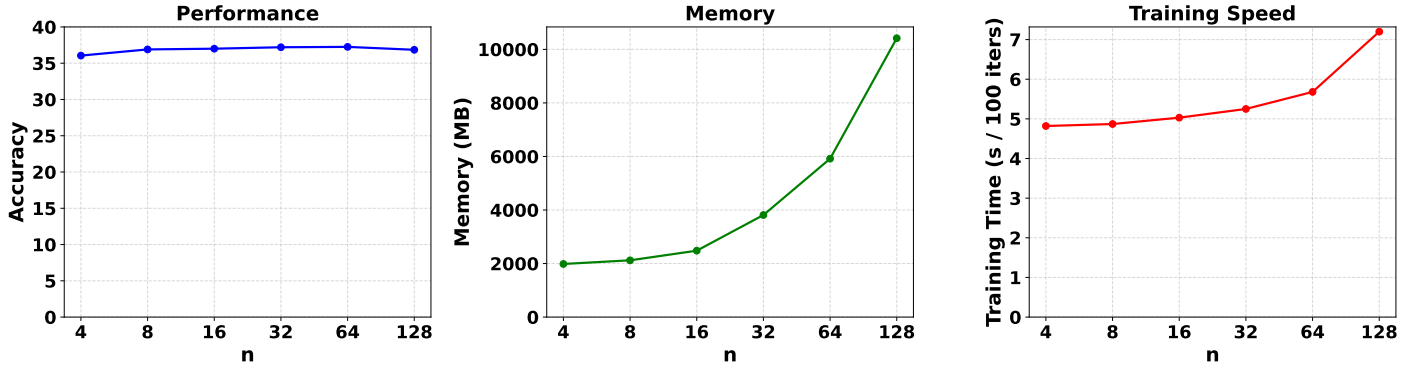


Figure 3: Performance, memory consumption, and training speed of Flexformer _{n} on the ListOps dataset for Long Sequence Classification under different values of the sampling frequency hyperparameter n . As n increases, both memory usage and training time also increase. On ListOps, competitive performance can be achieved even with a very small n . In the main experiments, we use $n = 64$.